

Student Guide — senior / module-07-eleven-multiplication

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Ekādhikena Pūrveṇa* — _____
- *sūtra* — _____
- *Anurūpyena* — _____

Day-by-day notes

Day 1 — The Hook — What Is the Trick?

What I learned:

One question I have:

Day 2 — The Carry Case

What I learned:

One question I have:

Day 3 — Why It Works — The Algebra

What I learned:

One question I have:

Day 4 — 3-Digit $\times$ 11

What I learned:

One question I have:

Day 5 — Speed Day

What I learned:

One question I have:

Day 6 — Anurūpyena — $\times 12$ to $\times 19$

What I learned:

One question I have:

Day 7 — Limits of the Method

What I learned:

One question I have:

Day 8 — Tirthaji and Historical Context

What I learned:

One question I have:

Day 9 — Mixed Practice

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Hook — What Is the Trick?

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 23×11 , 35×11 mentally using the split-and-sum method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. “by one more than the previous” — sūtra 1 in Tirthaji)

Primary source (today’s reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Compute 23×11 silently in your head.** Pause. Show of hands for who got 253. Then write $23 \times 11 = 253$ on the board.
2. **Reveal the trick.** Split the digits of 23 (gap in the middle): **2 _ 3**. Add the digits in the middle: $2 + 3 = 5$. Fill the gap: **2 5 3** = 253. Done.
3. **Try together.** $35 \times 11 = 3_ (3+5) _ 5 = 385$. $24 \times 11 = 264$. $71 \times 11 = 781$. $52 \times 11 = 572$. $80 \times 11 = 880$. Pace: 4 problems in 2 minutes.
4. **Tirthaji’s framing.** This method appears as an application of the sūtra *Ekādhikena Pūrveṇa* (“by one more than the previous”) in Bharati Krishna Tirthaji’s *Vedic Mathematics* (1965). Honest historical note: the manuscript was written 1911–1918; some scholars (Dani 1993) consider it a 20th-century mental-arithmetic system rather than strictly ancient mathematics.

5. **Independent practice (5 min).** Worksheet: 20 two-digit $\times$ 11 problems, no carry case. Pairs compete.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Hook — What Is the Trick?.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — The Carry Case

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 87×11 , 65×11 correctly using the carry rule.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *sūtra* (IAST: *sūtra*; lit. “aphorism / formula”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Try 87×11 .** Apply the rule: $8 _ (8+7) _ 7 = 8 _ 15 _ 7$. But 15 doesn’t fit in one slot. So *carry*: keep the 5 in the middle, add the 1 to the 8 $\rightarrow 9 \ 5 \ 7 = \mathbf{957}$.
2. **General rule extension.** When the middle sum ≥ 10 :
 - Write the units of the sum in the middle.
 - Carry the tens digit to the left, adding it to the existing left digit.
3. **Practice progression.**
 - $65 \times 11 = 6 _ (11) _ 5 \rightarrow 715$
 - $49 \times 11 = 4 _ (13) _ 9 \rightarrow 539$
 - $99 \times 11 = 9 _ (18) _ 9 \rightarrow 1089$
 - $84 \times 11 = 9 _ (12) _ 4 \rightarrow 924$
4. **Cross-check with standard long multiplication.** Have students verify two of the answers using the traditional algorithm. “*Same answer? Good. The Vedic method is just a different procedure for the same arithmetic.*”
5. **Limit recognition.** “*When might this be slower than long multiplication?*” For numbers with three or more digits where carries cascade, the saving shrinks. Tomorrow we extend to 3-digit and meet the algebraic proof.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces The Carry Case.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Why It Works — The Algebra

Module: ×11 Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive $(10a+b) \times 11 = 100a + 10(a+b) + b$ and explain.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: *Anurūpyena*; lit. “proportionately” — *sūtra* used for ×12, ×13 extensions”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Write any 2-digit number as $10a + b$.** Examples: $23 = 10 \cdot 2 + 3$ ($a=2$, $b=3$); $87 = 10 \cdot 8 + 7$.

2. **Distribute multiplication.**

$$\begin{aligned}(10a + b) \times 11 &= (10a + b)(10 + 1) \\ &= (10a + b) \cdot 10 + (10a + b) \cdot 1 \\ &= 100a + 10b + 10a + b \\ &= 100a + 10(a + b) + b\end{aligned}$$

3. **Read the final line in plain language.**

- $100a$ → hundreds digit is a
- $10(a+b)$ → tens digit is $a+b$ (the digit sum!)
- $+b$ → units digit is b

This IS the trick. The algebra spells out the procedure.

4. **Verify on 23.** $100 \cdot 2 + 10 \cdot (2+3) + 3 = 200 + 50 + 3 = 253$. ✓
5. **Connection to the carry case.** When $a+b \geq 10$, $10(a+b)$ becomes $10(a+b) = 100 + 10 \cdot (a+b-10)$, pushing the overflow into the hundreds place. The algebra automatically handles the carry.
6. **Independent task.** Each student writes the proof in their own words for a homework explainer to a younger student.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — 3-Digit $\times$ 11

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute $234 \times 11 = 2_5_7_4$ using the extended split method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tīrthajī”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Generalise to 3-digit.** For abc ($= 100a + 10b + c$) $\times 11$: $(100a + 10b + c)(10 + 1) = 1000a + 100b + 10c + 100a + 10b + c = 1000a + 100(a+b) + 10(b+c) + c$
2. **Procedure.** Split abc with two gaps: **a _ b _ c**. Fill gaps with adjacent sums: $a+b$, then $b+c$. Result: **a (a+b) (b+c) c**. Carry where needed.
3. **Worked examples.**
 - $234 \times 11 = 2_ (2+3)_ (3+4)_ 4 = 2574$
 - $567 \times 11 = 5_ (5+6)_ (6+7)_ 7 = 6237$ (carries: $11 \rightarrow$ carry 1 to left; $13 \rightarrow$ carry 1 to next left)
 - $729 \times 11 = 7_ (7+2)_ (2+9)_ 9 = 8019$
4. **Algebra check.** $234 = 100 \cdot 2 + 10 \cdot 3 + 4$. $\times 11 = 1000 \cdot 2 + 100 \cdot 5 + 10 \cdot 7 + 4 = 2574$. ✓
5. **Extension activity.** Each pair solves five 3-digit $\times 11$ problems, then verifies by long multiplication. Time-budget 8 minutes; share fastest correct-answer time.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces 3-Digit $\times 11$.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Speed Day

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: complete a 50-problem mental-math sprint in 5 minutes.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *sūtra* (IAST: *sūtra*; lit. “aphorism / formula”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Setup.** 50-problem mental-math sprint. Mixed 2- and 3-digit $\times 11$. Half no-carry, half carry. Five minutes on the clock.
2. **Solo run.** Silence. Pen-on-paper. Five minutes.
3. **Pair-mark.** Swap with partner; mark in two minutes.
4. **Analyse errors.** “*Most common error?*” (Forgetting to carry; transposing digits.) “*Fastest sub-strategy?*” Discuss.
5. **Standards comparison.** Equivalent traditional long-multiplication sprint would take ~ 15 minutes. Time-saving on this problem class: $\sim 3\times$. State the limit: this is for $\times 11$ specifically. Tomorrow we extend to $\times 12$ through $\times 19$.
6. **Personal best.** Each student records their score and target for next week.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Speed Day.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Anurūpyena — $\times 12$ to $\times 19$

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: extend the method to $\times 12$ and $\times 13$ using one-more-digit adjustments.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”

- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Move beyond $\times 11$.** Tirthaji’s *Anurūpyena* (“proportionately”) sūtra extends the digit-sum idea to $\times 12$ through $\times 19$. Procedure for $\times 12$:
 - Write any number N .
 - Compute: $N \times 12 = (N \times 10) + (N \times 2)$ — easy mental sum.
 - Equivalent digit-procedure: split-double-add.
2. **Worked example, 23×12 .**
 - $23 \times 10 = 230$
 - $23 \times 2 = 46$
 - Sum: 276
 - OR digit-wise: $2_3 \rightarrow 2_ (2 \cdot 1 + 3) _ (3 \cdot 1)$ wait — this gets tangled. The cleanest mental version is the $(\times 10) + (\times k)$ split.
3. **General $\times(10+k)$ rule.** $N \times (10+k) = 10N + kN$. The *Anurūpyena* framing emphasises the proportional scaling.
4. **Practice.** Compute mentally:
 - $14 \times 13 = 140 + 42 = 182$
 - $25 \times 15 = 250 + 125 = 375$
 - $32 \times 17 = 320 + 224 = 544$
5. **Honest historical note.** The $(10+k)$ decomposition isn’t unique to Tirthaji — Western mental-math systems describe the same trick. Tirthaji’s contribution is grouping it under a Sanskrit sūtra-system.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces *Anurūpyena* — $\times 12$ to $\times 19$.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Limits of the Method

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recognise when traditional long multiplication is faster than the *sūtra* method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: *Ekādhikena Pūrveṇa*; lit. ““by one more than the previous” — *sūtra* 1 in *Tīrthajī*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Pose the question.** “*When IS the $\times 11$ (or near-base) method genuinely faster than traditional long multiplication? When is it the SAME or SLOWER?*”
2. **Categorise problem types.**
 - 2-digit $\times 11$ — much faster.
 - 3-digit $\times 11$ — somewhat faster.
 - 2-digit $\times 12$ – 19 — comparable.
 - 2-digit $\times 23$ (far from base) — long multiplication is faster.
 - Long multi-digit products — long multiplication or scientific notation is best.
3. **Honest framing.** *Tīrthajī’s sūtras are not a universal speed-up. They are a toolbox of mental shortcuts for specific patterns.* The skill is recognising when to use which tool.

4. **Reference.** Modern mathematicians (S.G. Dani, 1993; Kim Plofker, *Mathematics in India*, 2009) have written critically about the “Vedic” claim and pedagogically about the methods. Read Dani’s *Frontline* article excerpt aloud (one paragraph).
5. **Reflection.** Each student writes 3 sentences: which method-types are useful, what limits are, and where standard arithmetic remains best.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Limits of the Method.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Tirthaji and Historical Context

Module: ×11 Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain who tirthaji was and why scholars debate the “vedic” attribution.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *sūtra* (IAST: *sūtra*; lit. “aphorism / formula”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Who was Tirthaji?** Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha, Puri. Composed the manuscript that became *Vedic Mathematics* during 1911–1918. Published posthumously 1965.
2. **Tirthaji’s claim.** The 16 sūtras + 13 sub-sūtras were “reconstructed” from the *Parīśiṣṭa of the Atharva Veda*. The original manuscript with sūtras has never been independently located.
3. **Scholarly response.** S.G. Dani (Tata Institute of Fundamental Research), in *Frontline* (1993), examined the claim: the sūtras as published do NOT appear in extant Atharva Veda Parīśiṣṭa manuscripts. Most academic mathematicians treat Tirthaji’s system as a 20th-century mental-arithmetic synthesis, drawing on traditional and modern sources, NOT strictly ancient mathematics.
4. **Pedagogical value.** The methods themselves work and teach number sense effectively. The *historical claim* is separable from the *pedagogical method*. Many curricula in India today teach the methods while updating the historical claim.
5. **Reading.** Hand out a one-page excerpt from Dani’s article and Tirthaji’s preface side-by-side. Pairs discuss: “*What does each author claim? What evidence does each provide?*”

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Tirthaji and Historical Context.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Mixed Practice

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve a mixed problem set using best-fit method per problem.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: *Anurūpyena*; lit. “proportionately” — *sūtra* used for $\times 12$, $\times 13$ extensions”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed-method practice set.** 25 problems, 15 minutes. Each problem is either:
 - Easy $\times 11$ (no carry)
 - Tricky $\times 11$ (with carry)
 - $\times 12$ or $\times 13$ (*Anurūpyena*)
 - Standard 2-digit $\times$ 2-digit (use any method)
2. **Self-routing.** Students decide WHICH method to apply per problem, justified silently. Speed and accuracy both matter.
3. **Pair-mark + discuss.** Two minutes per pair to compare strategies. “*Did you both pick the same method? When did you disagree?*”

4. **Whole-class harvest.** Two students share one problem each where method choice mattered.
5. **Reflection (3 sentences).** Each student writes which method-type they're confident in, which they want more practice on, which they avoid.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed Practice.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Assessment

Module: ×11 Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: take the summative quiz and reflect on which method works best.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: *Ekādhikena Pūrveṇa*; lit. ““by one more than the previous” — sūtra 1 in *Tīrthajī*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Method-comparison reflection (10 min).** Each student writes a one-paragraph response: “*Compare the Tirthaji $\times 11$ method with traditional long multiplication. When is each preferable? What does Tirthaji’s system teach you that standard arithmetic doesn’t?*”
2. **Honest history check (5 min).** “*In 2-3 sentences, summarise the historical debate about ‘Vedic Mathematics.’ What’s claimed? What does scholarship say?*”
3. **Summative quiz (25 min).** Mix of computation, algebraic proof, and conceptual questions. See `quizzes/summative.md`.
4. **Project share (5 min).** Each student shows the method-comparison poster they built (`assessments/project-brief.md`).
5. **Module close.** “*What’s one number-sense habit from this module that you’ll keep using?*” Each student shares one sentence.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Final Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain ×11 Multiplication — Ekādhikena Pūrveṇa to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare ×11 Multiplication — Ekādhikena Pūrveṇa with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about ×11 Multiplication — Ekādhikena Pūrveṇa? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Hook — What Is the Trick?** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — The Carry Case** · Write 2 sentences about today's concept. (~10 min)

3. **Day 3 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — 3-Digit $\times$ 11** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Speed Day** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Anurūpyena — $\times 12$ to $\times 19$** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Limits of the Method** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Tirthaji and Historical Context** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.